

1. What is SQL and What are its Uses?

SQL (Structured Query Language) is a domain-specific language used to store, retrieve, manipulate, and manage data in a Relational Database Management System (RDBMS).

SQL is mainly used for handling structured data, where data is organized into tables consisting of rows and columns. SQL is generally a non-case-sensitive language.

Uses of SQL:

- **SQL can retrieve data from a database.**
- **SQL can insert new data into a database.**
- **SQL can create new databases.**
- **SQL can create new tables in databases.**
- **SQL can update existing data in databases.**
- **SQL can delete data from databases.**
- **SQL can perform queries related to databases.**
- **SQL can create and manage database objects such as tables, views, and indexes.**
- **SQL can set permissions on tables, procedures, views, and other database objects.**

SQL Standards:

SQL is an American National Standards Institute (ANSI) and International Organization for Standardization (ISO) standard.

Although SQL is an ANSI/ISO standard, different database management systems may have different versions and extensions of SQL. However, most of them support the major SQL commands, such as SELECT, UPDATE, DELETE, INSERT, and WHERE, in a similar manner.

2. Different Commands of SQL, Their Uses and Syntax

The following are some of the commonly used SQL commands:

1. SELECT Command Use:

The SELECT command is used to retrieve data from a database table.

Syntax:

```
SELECT column1, column2, ... FROM table_name;
```

To select all columns:

SELECT *

```
FROM table_name;
```

2. UPDATE Command

Use:

The UPDATE command is used to modify existing data in a database table.

Syntax:

```
UPDATE table_name  
SET column1 = value1, column2 = value2, ... WHERE condition;
```

3. DELETE Command

Use:

The DELETE command is used to delete existing records from a database table.

Syntax:

```
DELETE FROM table_name WHERE condition;
```

4. INSERT INTO Command Use:

The INSERT INTO command is used to insert new records into a database table.

Syntax:

```
INSERT INTO table_name (column1, column2, column3, ...) VALUES (value1,  
value2, value3, ...);
```

5. CREATE DATABASE Command

Use:

The CREATE DATABASE command is used to create a new database.

Syntax:

```
CREATE DATABASE [IF NOT EXISTS] database_name [CHARACTER SET charset_name]  
[COLLATE collation_name];
```

6. ALTER DATABASE Command

Use:

The ALTER DATABASE command is used to modify the properties or settings of an existing database.

Syntax:

```
ALTER DATABASE [database_name] [DEFAULT] CHARACTER SET charset_name [DEFAULT]  
COLLATE collation_name;
```

7. CREATE TABLE Command Use:

The CREATE TABLE command is used to create a new table in a database.

Syntax:

```
CREATE TABLE table_name (  
column1_name data_type(size) [constraints],
```

```
column2_name data_type(size) [constraints],  
...  
[table_constraints]  
);
```

8. ALTER TABLE Command Use:

The ALTER TABLE command is used to modify the structure of an existing table. It can be used to add, modify, or remove columns and constraints.

Syntax:

```
ALTER TABLE table_name action1,  
action2,  
...;
```

9. DROP TABLE Command

10.

Use:

The DROP TABLE command is used to permanently delete a table and all its data from a database.

Syntax:

```
DROP TABLE [IF EXISTS] table_name;
```

11. CREATE INDEX Command

Use:

The CREATE INDEX command is used to create an index on one or more columns. An index can improve the speed of data retrieval operations.

Syntax:

```
CREATE [UNIQUE | FULLTEXT | SPATIAL] INDEX index_name  
ON table_name (column1 [ASC | DESC], column2 [ASC | DESC], ...);
```

12.DROP INDEX Command Use:

The DROP INDEX command is used to remove an existing index from a table.

Syntax:

```
DROP INDEX index_name ON table_name;
```

MySQL Installation:

Step 1 – Download MySQL Installer

Select the Windows MSI Installer download.

[General Availability \(GA\) Releases](#) [Archives](#) [Info](#)

MySQL Installer 8.0.35

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:
8.0.35

Select Operating System:
Microsoft Windows

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.35.0.msi)	8.0.35	2.1M	Download
MD5: 214df2ccdf83eb5edc6ca7c115792406 Signature			
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.35.0.msi)	8.0.35	288.6M	Download
MD5: 2cfda448a2971b6b5323775ef9e8d012 Signature			

Step 2 – Start the Download

Select “No thanks, just start my download.”

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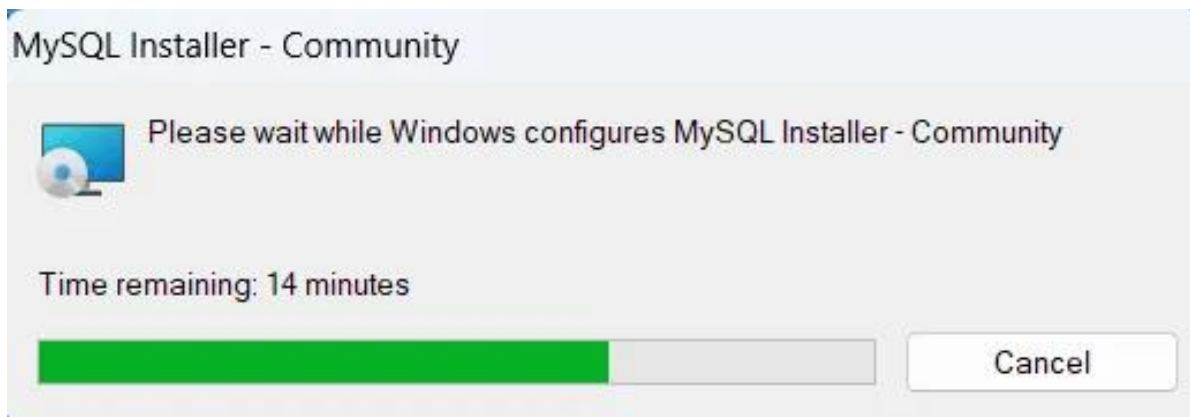
for an Oracle Web account

MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

[No thanks, just start my download.](#)

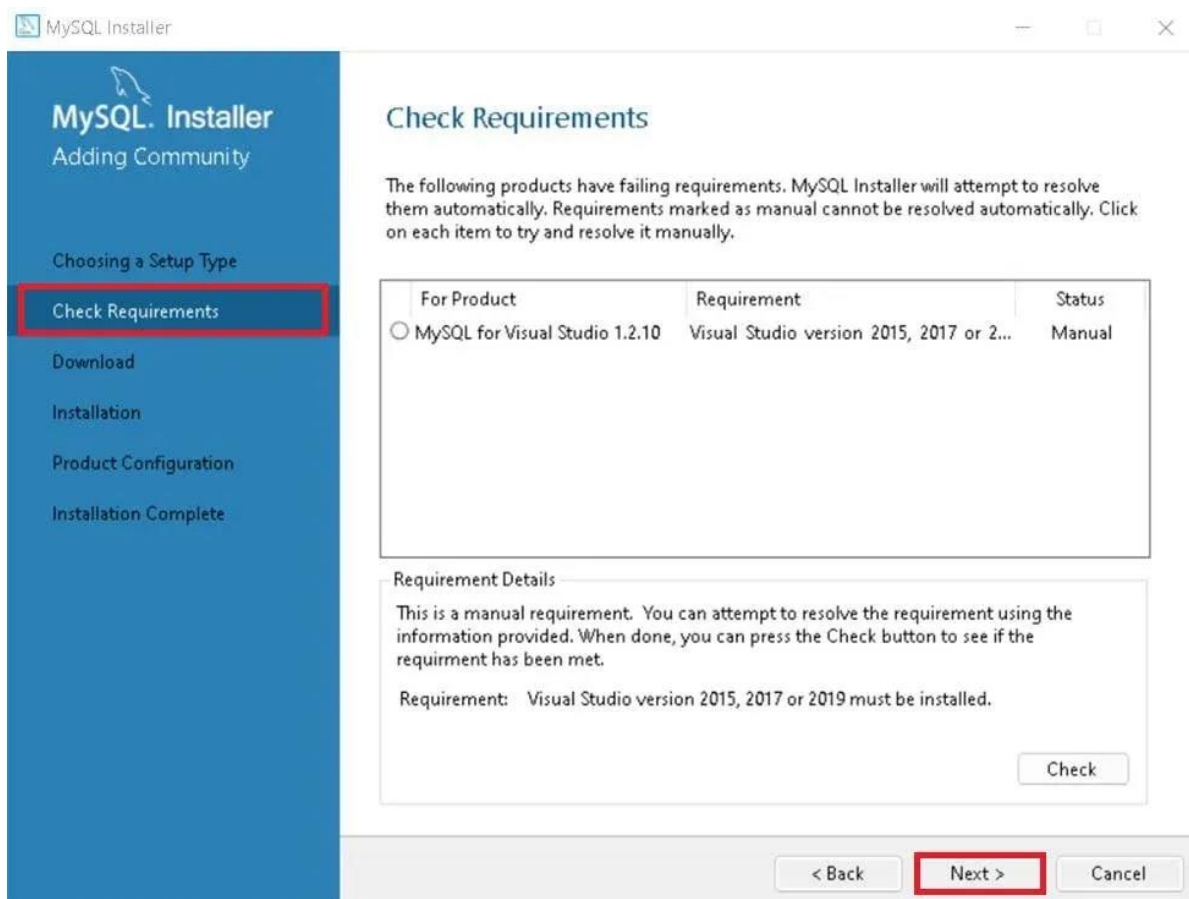
Step 3 – Wait for the Installer

Wait while Windows configures MySQL Installer.



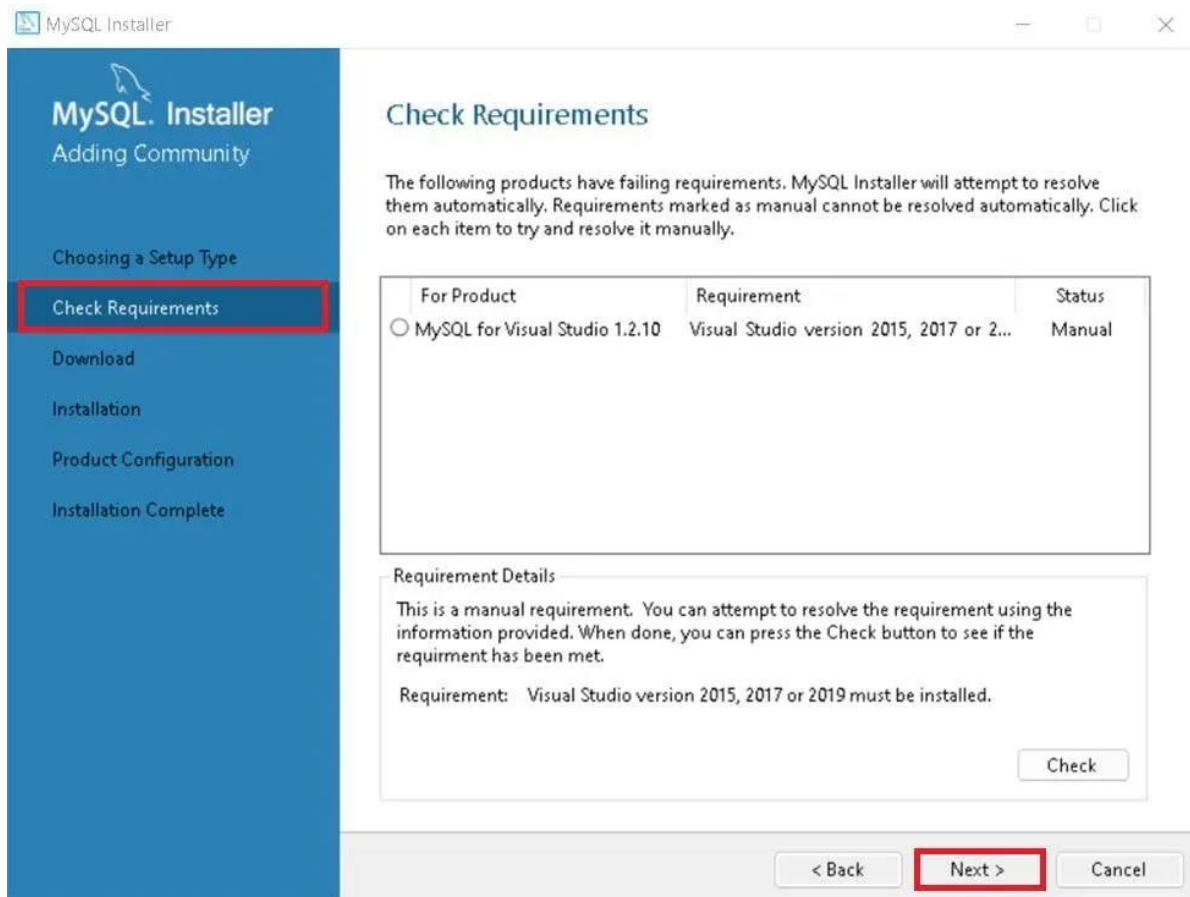
Step 4 – Check Requirements

Check the listed requirements and continue with Next.



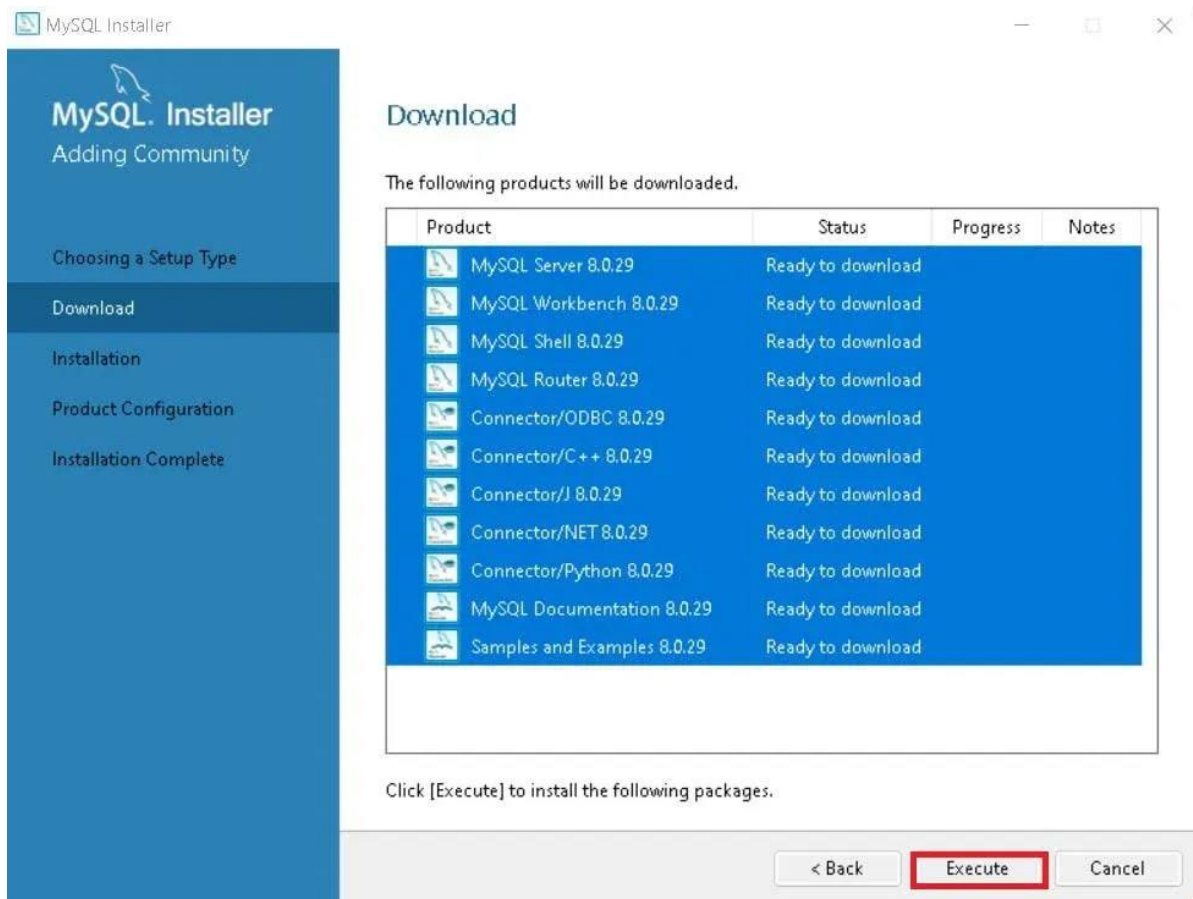
Step 5 – Continue

Continue to the Download section.



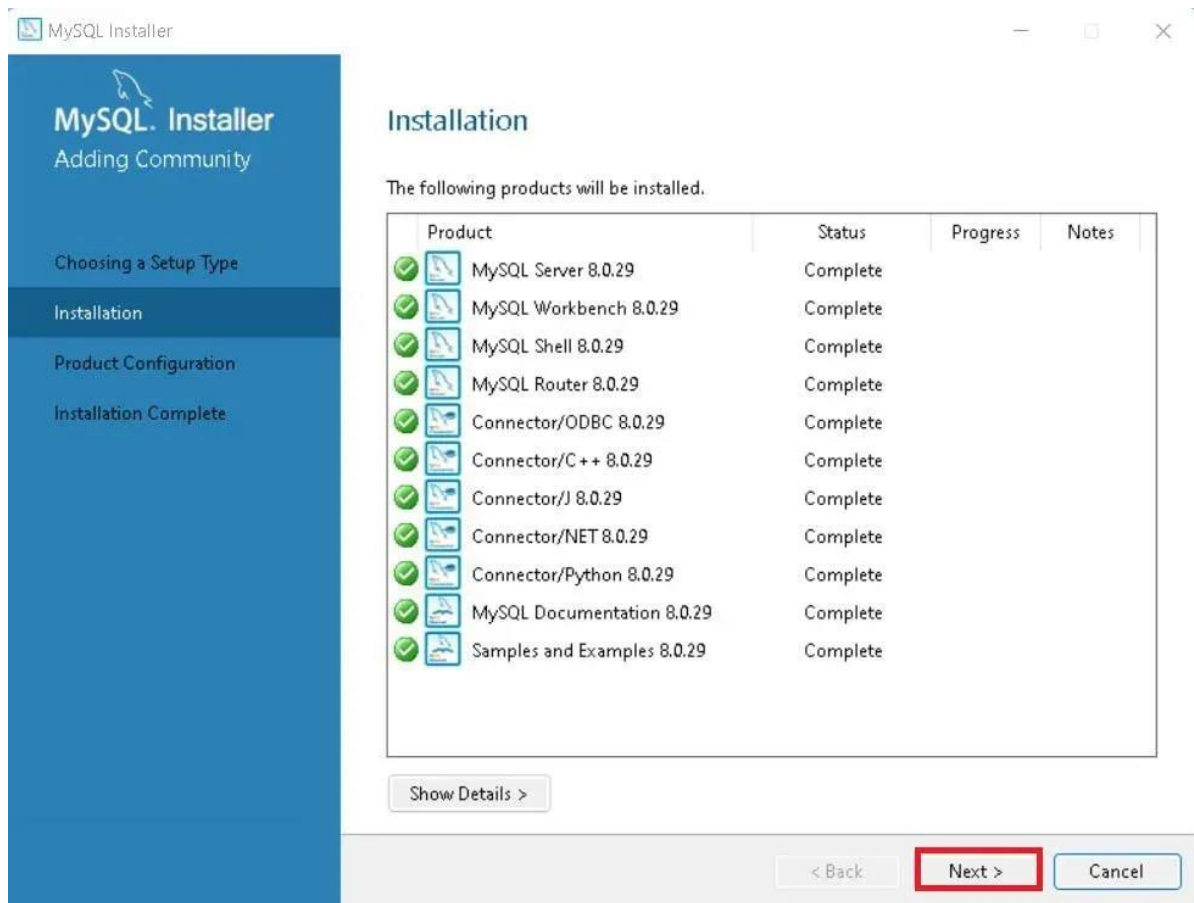
Step 6 – Download Packages

Click Execute to download and install the listed packages.



Step 7 – Installation

Wait until the listed products show Complete, then click Next.



Step 8 – Accounts and Roles

Enter the MySQL Root Password and repeat it.

MySQL Installer

MySQL Server 8.0.29

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: **Strong**

MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role
-----------------	------	-----------

Add User

Edit User

Delete

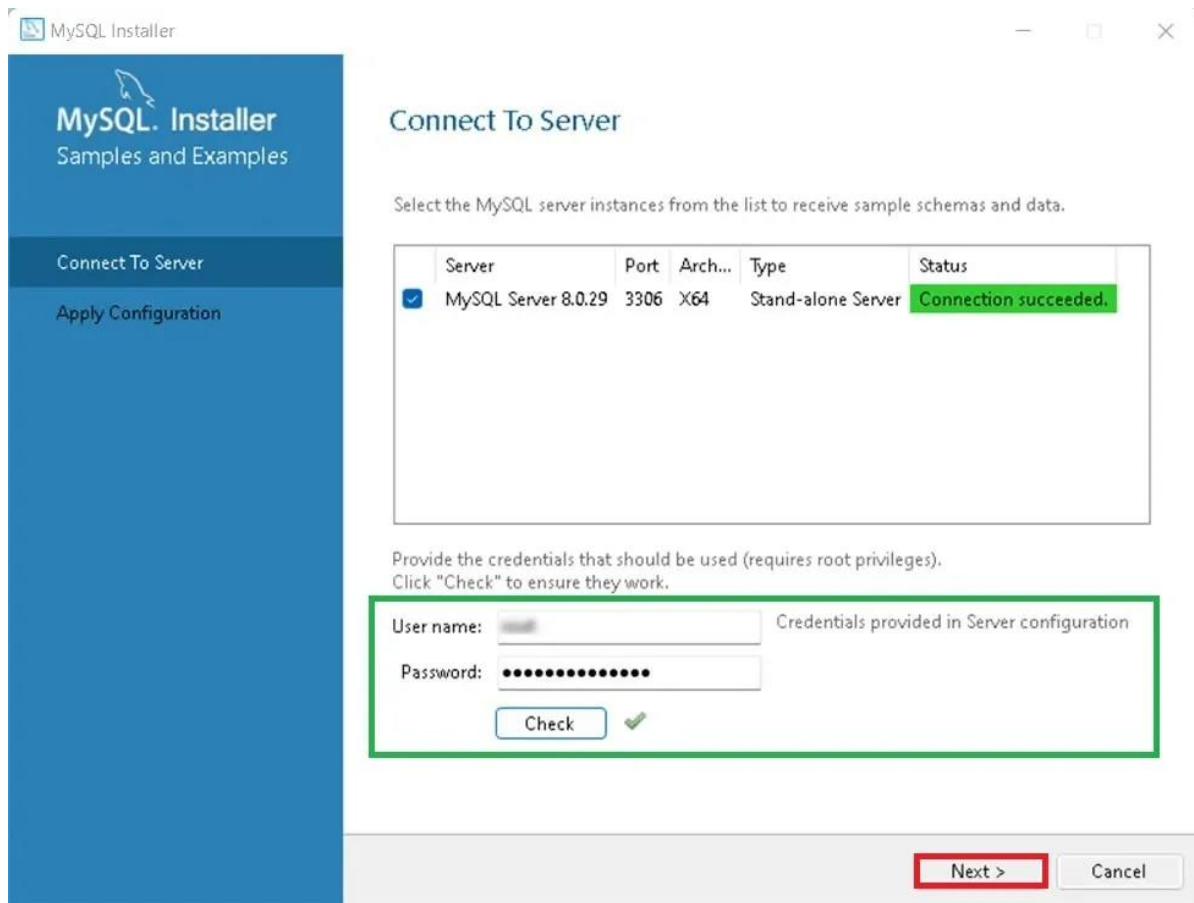
< Back

Next >

Cancel

Step 9 – Connect to Server

Enter the credentials and click Check until the connection succeeds.



Step 10 – Installation Complete

Click Finish to complete the installation

MySQL. Installer

Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

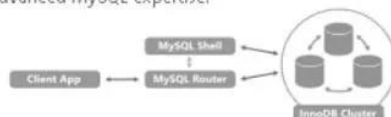
Installation Complete

The installation procedure has been completed.

Copy Log to Clipboard

- ☒ Start MySQL Workbench after setup
- ☒ Start MySQL Shell after setup

The MySQL Shell is an advanced MySQL client application that can be used to work with single MySQL Server instances. Further, it can be used to create and manage InnoDB Cluster, an integrated solution for high availability and scalability of MySQL databases, without requiring advanced MySQL expertise.



Refer to the following links for documentation, tutorials and examples on MySQL Shell:

[MySQL Shell Documentation](#)

[Setting up a Real World Cluster Blog](#)

[The All New MySQL InnoDB ReplicaSet Blog](#)

[Changing Cluster Options Live Blog](#)

Finish